(2) In[•]:=

```
Exit[]
```

Ramanujan Challenge 2.4: the RISC challenge

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Using the RISC packages:

(2) In[1]:=

```
<< Sigma.m;
<< HarmonicSums.m;
<< EvaluateMultiSums.m;
```

```
Sigma
 - A summation package by Carsten Schneider – © RISC – V 2.94 (June 20, 2026) click for Help
```

```
HarmonicSums by Jakob Ablinger – © RISC – Version 1.0 (30 / 03 / 21) Help
```

```
EvaluateMultiSums by Carsten Schneider (still under development ...) –
 © RISC – V 0.996 (March 11, 2022)
```

we will simplify the double sum

(2) In[4]:=

```
mySum = SigmaSum[SigmaSum[SigmaBinomial[m, k]^2 × S[1, k]^2, {k, 0, m}] / (m + 1)^2 /
    SigmaBinomial[2 m, m], {m, 0, Infinity}]
```

(2) Out[4]=

$$\sum_{m=0}^{\infty} \frac{\sum_{k=0}^{m} \left(\binom{m}{k}\right)^2 S[1, k]^2}{(1+m)^2 \binom{2m}{m}}$$

to

$$20\,\text{li4half} + \frac{5\,\text{ln2}^4}{6} + 10\,\text{z2} - \frac{65\,\text{z2}^2}{9} - \text{ln2}^2\left(12 + 5\,\text{z2}\right) + \frac{\text{z3}}{2} + \text{ln2}\left(-16 + \frac{35\,\text{z3}}{2}\right)$$

Note : ln2 = Log[2], z2 = Zeta[2] and li4half = PolyLog[4, 1/2]

Here will perform two steps: In a first round we simplify automatically the double sum in terms of known constants and two infinite binomial sums. In the second step also these unsimplified binomial sums are reduced to known constants; this can be done also automatically, but an extra step has to be carried out by a new function definition.

We remark that this machinery is rather general and allows one to tackle similar problems fully automatically using the summation machinery Sigma and HarmonicSums.

A representation in terms of infinite binomials sums

First we use our symbolic summation machinery and get

(2) In[5]:=

```
resInfiniteBinSum = EvaluateMultiSum[mySum, {}, {}, {}] /. LinInf → Infinity
```

```
Loading Relations...
```

(2) Out[5]=

$$20\,\text{li4half} + \frac{5\,\ln 2^4}{6} + \ln 2^2\left(-12 - 5\,z2\right) - \frac{38\,z2^2}{5} + \frac{z3}{2} + \ln 2\left(-16 + \frac{35\,z3}{2}\right) + z2\left(10 + 3\left(\sum_{i_1=1}^{\infty}{}^{\cdot}\,\frac{(i_1!.)^2}{i_1^2\,(2\,i_1)!.}\right)\right) - 3\left(\sum_{i_1=1}^{\infty}{}^{\cdot}\,\frac{(i_1!.)^2\,\text{S}[2,\,i_1]}{i_1^2\,(2\,i_1)!.}\right)$$

Details of the calculation can be found in the next subsection.

Details (with a CA proof)

We first look at the inner sum and afterwards turn to the outermost sum. For further details of the underlying machinery we refer to

C. Schneider Symbolic Summation Assists Combinatorics. Sem. Lothar. Combin. 56, pp. 1-36. 2007. ISSN 1286-4889. Article B56b.

C. Schneider Term Algebras, Canonical Representations and Difference Ring Theory for Symbolic Summation. In: Anti-Differentiation and the Calculation of Feynman Amplitudes, J. Blümlein and C. Schneider (ed.), Texts and Monographs in Symbolic Computuation , pp. 423-485. 2021.

J. Ablinger, J. Blümlein and C. Schneider. Harmonic Sums and Polylogarithms Generated by Cyclotomic Polynomials. J. Math. Phys. 52 (2011) 102301. arXiv:1105.6063 [math-ph].

Dealing with the inner sum

To simplify the inner sum

(2) In[6]:=

```
innerSum = mySum[[1, 3]]
```

(2) Out[6]=

$$\sum_{k=0}^{m} \left(\binom{m}{k}\right)^2 S[1, k]^2$$

we will utilize the Sigma summation spiral: (1) compute a recurrence via Zeilberger's creative telescoping paradigm, (2) solve the recurrence in terms of indefinite nested sums and (3) combine the solutions to obtain a closed form.

Computing a recurrence via Zeilberger's creative telescoping paradigm

We can compute a recurrence for SUM[m]=innerSum with Sigma as follows:

(2) In[7]:=

```
rec = GenerateRecurrence[innerSum][[1]]
```

(2) Out[7]=

$$\begin{aligned}
&8\,(1+m)\,(1+2\,m)^3\,(735+1621\,m+1332\,m^2+480\,m^3+64\,m^4)\,\mathrm{SUM}[m]- \\
&\quad 4\,(4533+33\,460\,m+106\,442\,m^2+183\,201\,m^3+186\,452\,m^4+115\,708\,m^5+43\,056\,m^6+8832\,m^7+768\,m^8) \\
&\quad \mathrm{SUM}[1+m]+2\,(2+m)\,(2616+16\,084\,m+42\,775\,m^2+58\,679\,m^3+44\,610\,m^4+18\,968\,m^5+4224\,m^6+384\,m^7) \\
&\quad \mathrm{SUM}[2+m]-(2+m)\,(3+m)^3\,(30+141\,m+276\,m^2+224\,m^3+64\,m^4)\,\mathrm{SUM}[3+m]== \\
&-3\,(1854+11\,574\,m+26\,171\,m^2+28\,761\,m^3+16\,660\,m^4+4896\,m^5+576\,m^6)
\end{aligned}$$

To verify the correctness of the recurrence, one has to look at the summand recurrence that one can obtain (as a subresult of the above calculation) as follows:

(2) In[8]:=

```
{c0, c1, c2, c3, g} = CreativeTelescoping[innerSum][1]
```

(2) Out[8]=

$$\Big\{8\,(1+m)\,(1+2\,m)^3\,\left(735+1621\,m+1332\,m^2+480\,m^3+64\,m^4\right),$$
$$-4\,\left(4533+33\,460\,m+106\,442\,m^2+183\,201\,m^3+186\,452\,m^4+115\,708\,m^5+43\,056\,m^6+8832\,m^7+768\,m^8\right),$$
$$2\,(2+m)\,\left(2616+16\,084\,m+42\,775\,m^2+58\,679\,m^3+44\,610\,m^4+18\,968\,m^5+4224\,m^6+384\,m^7\right),$$
$$-\left((2+m)\,(3+m)^3\,\left(30+141\,m+276\,m^2+224\,m^3+64\,m^4\right)\right),\ \left(\binom{m}{k}^{\cdot}\right)^2$$
$$\Big(\big(22\,248-102\,960\,k+199\,014\,k^2-179\,358\,k^3+74\,970\,k^4-11\,760\,k^5+205\,632\,m-780\,836\,k\,m+1\,228\,174$$
$$k^2\,m-899\,954\,k^3\,m+303\,522\,k^4\,m-37\,696\,k^5\,m+803\,022\,m^2-2\,514\,858\,k\,m^2+3\,224\,490\,k^2\,m^2-$$
$$1\,903\,866\,k^3\,m^2+505\,292\,k^4\,m^2-47\,248\,k^5\,m^2+1\,772\,046\,m^3-4\,552\,430\,k\,m^3+4\,713\,150\,k^2\,m^3-$$
$$2\,197\,970\,k^3\,m^3+442\,024\,k^4\,m^3-28\,992\,k^5\,m^3+2\,469\,879\,m^4-5\,134\,218\,k\,m^4+4\,203\,590\,k^2\,m^4-$$
$$1\,494\,804\,k^3\,m^4+213\,984\,k^4\,m^4-8704\,k^5\,m^4+2\,285\,991\,m^5-3\,757\,334\,k\,m^5+2\,346\,950\,k^2\,m^5-$$
$$598\,768\,k^3\,m^5+54\,272\,k^4\,m^5-1024\,k^5\,m^5+1\,429\,119\,m^6-1\,790\,196\,k\,m^6+802\,360\,k^2\,m^6-$$
$$130\,816\,k^3\,m^6+5632\,k^4\,m^6+597\,843\,m^7-536\,880\,k\,m^7+153\,792\,k^2\,m^7-12\,032\,k^3\,m^7+160\,572\,m^8-$$
$$92\,160\,k\,m^8+12\,672\,k^2\,m^8+25\,056\,m^9-6912\,k\,m^9+1728\,m^{10}\big)\Big/\big((-2+k-m)^2\,(-1+k-m)^2\big)-$$
$$\Big(2\,(1+m)\,\big(22\,248\,k-77\,868\,k^2+83\,082\,k^3-36\,750\,k^4+5880\,k^5+183\,384\,k\,m-537\,012\,k^2\,m+$$
$$489\,254\,k^3\,m-183\,950\,k^4\,m+24\,728\,k^5\,m+619\,638\,k\,m^2-1\,523\,775\,k^2\,m^2+1\,156\,794\,k^3\,m^2-$$
$$352\,340\,k^4\,m^2+36\,592\,k^5\,m^2+1\,152\,408\,k\,m^3-2\,349\,630\,k^2\,m^3+1\,445\,118\,k^3\,m^3-340\,160\,k^4\,m^3+$$
$$25\,152\,k^5\,m^3+1\,317\,471\,k\,m^4-2\,173\,395\,k^2\,m^4+1\,040\,472\,k^3\,m^4-176\,960\,k^4\,m^4+$$
$$8192\,k^5\,m^4+968\,520\,k\,m^5-1\,243\,368\,k^2\,m^5+434\,560\,k^3\,m^5-47\,360\,k^4\,m^5+1024\,k^5\,m^5+$$
$$460\,599\,k\,m^6-431\,760\,k^2\,m^6+97\,920\,k^3\,m^6-5120\,k^4\,m^6+137\,244\,k\,m^7-83\,520\,k^2\,m^7+$$
$$9216\,k^3\,m^7+23\,328\,k\,m^8-6912\,k^2\,m^8+1728\,k\,m^9\big)\,S[1,\,k]\Big)\Big/\big((1-k+m)^2\,(2-k+m)^2\big)-$$
$$\Big(k^2\,(1+m)\,\big(-200\,232+608\,472\,k-633\,564\,k^2+300\,696\,k^3-67\,620\,k^4+5880\,k^5-1\,783\,944\,m+$$
$$5\,026\,788\,k\,m-4\,878\,360\,k^2\,m+2\,158\,996\,k^3\,m-451\,952\,k^4\,m+36\,488\,k^5\,m-6\,699\,294\,m^2+$$
$$17\,409\,372\,k\,m^2-15\,536\,174\,k^2\,m^2+6\,280\,440\,k^3\,m^2-1\,190\,236\,k^4\,m^2+86\,048\,k^5\,m^2-$$
$$14\,272\,884\,m^3+33\,667\,588\,k\,m^3-27\,060\,396\,k^2\,m^3+9\,715\,892\,k^3\,m^3-1\,604\,128\,k^4\,m^3+$$
$$98\,336\,k^5\,m^3-19\,391\,325\,m^4+40\,675\,262\,k\,m^4-28\,714\,374\,k^2\,m^4+8\,853\,912\,k^3\,m^4-$$
$$1\,213\,552\,k^4\,m^4+58\,496\,k^5\,m^4-17\,773\,914\,m^5+32\,331\,542\,k\,m^5-19\,421\,012\,k^2\,m^5+$$
$$4\,918\,384\,k^3\,m^5-521\,920\,k^4\,m^5+17\,408\,k^5\,m^5-11\,273\,982\,m^6+17\,224\,678\,k\,m^6-$$
$$8\,428\,648\,k^2\,m^6+1\,639\,360\,k^3\,m^6-119\,296\,k^4\,m^6+2048\,k^5\,m^6-4\,967\,310\,m^7+6\,100\,290\,k\,m^7-$$
$$2\,274\,976\,k^2\,m^7+301\,568\,k^3\,m^7-11\,264\,k^4\,m^7-1\,494\,015\,m^8+1\,379\,304\,k\,m^8-347\,904\,k^2\,m^8+$$
$$23\,552\,k^3\,m^8-292\,764\,m^9+180\,288\,k\,m^9-23\,040\,k^2\,m^9-33\,696\,m^{10}+10\,368\,k\,m^{10}-1728\,m^{11}\big)$$
$$S[1,\,k]^2\Big)\Big/\big((-3+k-m)^2\,(-2+k-m)^2\,(-1+k-m)^2\big)\Big\}$$

More precisely, if we denote summand of the inner sum by f[m,k]

(2) In[9]:=

```
f = innerSum[1]
```

(2) Out[9]=

$$\left(\binom{m}{k}^{\cdot}\right)^2 S[1,\,k]^2$$

the following summand recurrence

c0 f[m, k] + c1 f[m+ 1, k] + c2 f[m + 2, k] + c3 f[m + 3, k] == g[m, k + 1] - g[m, k] (*)

holds for all k and m with 0 <= k <= m and m >= 0. This can be checked as follows:

(2) In[10]:=

```
c0 f + c1 NShift[f, m] + c2 NShift[f, m, 2] + c3 NShift[f, m, 3] - (NShift[g, k] - g) // Together
```

(2) Out[10]=

```
0
```

Finally, summing the summand recurrence (*) over k from 0 to m yields the above recurrence.

Solving the recurrence and combining the solutions

Next, we solve the recurrence with the following Sigma-call:

(2) In[11]:=

```
recSol = SolveRecurrence[rec, SUM[m]]
```

(2) Out[11]=

$$\left\{\left\{0, \frac{(2\,m)!.}{(m!.)^2}\right\}, \left\{0, \frac{(2\,m)!.\left(3\left(\sum_{\iota_1=1}^{m}\cdot \frac{1}{\iota_1}\right)-2\left(\sum_{\iota_1=1}^{m}\cdot \frac{1}{-1+2\,\iota_1}\right)\right)}{(m!.)^2}\right\},\right.$$

$$\left\{0, \frac{1}{(m!.)^2}(2\,m)!.\left(\frac{1}{2}\left(\sum_{\iota_1=1}^{m}\cdot \frac{1}{\iota_1^2}\right)-\frac{9}{2}\left(\sum_{\iota_1=1}^{m}\cdot \frac{1}{\iota_1}\right)^2+2\left(\sum_{\iota_1=1}^{m}\cdot \frac{1}{\left(-1+2\,\iota_1\right)^2}\right)+6\left(\sum_{\iota_1=1}^{m}\cdot \frac{1}{\iota_1}\right)\left(\sum_{\iota_1=1}^{m}\cdot \frac{1}{-1+2\,\iota_1}\right)-2\left(\sum_{\iota_1=1}^{m}\cdot \frac{1}{-1+2\,\iota_1}\right)^2\right)\right\},$$

$$\left.\left\{1, \frac{3\,(2\,m)!.\left(\sum_{\iota_1=1}^{m}\cdot \frac{(\iota_1!.)^2}{\iota_1^2\,(2\,\iota_1)!.}\right)}{(m!.)^2}\right\}\right\}$$

This means that

$$\frac{3\,(2\,m)!.\left(\sum_{\iota_1=1}^{m}\cdot \frac{(\iota_1!.)^2}{\iota_1^2\,(2\,\iota_1)!.}\right)}{(m!.)^2}$$

is a particular solution and

$$\frac{(2\,m)!.}{(m!.)^2},$$

$$\frac{(2\,m)!.\left(3\left(\sum_{\iota_1=1}^{m}\cdot \frac{1}{\iota_1}\right)-2\left(\sum_{\iota_1=1}^{m}\cdot \frac{1}{-1+2\,\iota_1}\right)\right)}{(m!.)^2},\ \frac{1}{(m!.)^2}(2\,m)!.$$

$$\left(\frac{1}{2}\left(\sum_{\iota_1=1}^{m}\cdot \frac{1}{\iota_1^2}\right)-\frac{9}{2}\left(\sum_{\iota_1=1}^{m}\cdot \frac{1}{\iota_1}\right)^2+2\left(\sum_{\iota_1=1}^{m}\cdot \frac{1}{\left(-1+2\,\iota_1\right)^2}\right)+6\left(\sum_{\iota_1=1}^{m}\cdot \frac{1}{\iota_1}\right)\left(\sum_{\iota_1=1}^{m}\cdot \frac{1}{-1+2\,\iota_1}\right)-2\left(\sum_{\iota_1=1}^{m}\cdot \frac{1}{-1+2\,\iota_1}\right)^2\right)$$

are three linearly independent solutions of the homogeneous version of the recurrence. Finally, we combine the three homogeneous solutions plus the particular solutions such that expression agrees with the evaluation of the innerSum at m=0,1,2:

(2) In[12]:=

```
clFormInnerSum = FindLinearCombination[recSol, innerSum, m, 3]
```

(2) Out[12]=

$$-\frac{(2\,m)!.\left(\sum_{i_1=1}^{m}\cdot\ \frac{1}{i_1^2}\right)}{4\,(m!.)^2}+\frac{9\,(2\,m)!.\left(\sum_{i_1=1}^{m}\cdot\ \frac{1}{i_1}\right)^2}{4\,(m!.)^2}-\frac{(2\,m)!.\left(\sum_{i_1=1}^{m}\cdot\ \frac{1}{(-1+2\,i_1)^2}\right)}{(m!.)^2}-$$

$$\frac{3\,(2\,m)!.\left(\sum_{i_1=1}^{m}\cdot\ \frac{1}{i_1}\right)\left(\sum_{i_1=1}^{m}\cdot\ \frac{1}{-1+2\,i_1}\right)}{(m!.)^2}+\frac{(2\,m)!.\left(\sum_{i_1=1}^{m}\cdot\ \frac{1}{-1+2\,i_1}\right)^2}{(m!.)^2}+\frac{3\,(2\,m)!.\left(\sum_{i_1=1}^{m}\cdot\ \frac{(i_1!.)^2}{i_1^2\,(2\,i_1)!.}\right)}{(m!.)^2}$$

Indeed, the first three values agree:

(2) In[13]:=

```
Table[clFormInnerSum - innerSum, {m, 0, 2}]
```

(2) Out[13]=

```
{0, 0, 0}
```

We check that also the found expression is a solution of the recurrence:

(2) In[14]:=

```
(rec[[1]] - rec[[2]]) /. SUM[A_] :> NShift[clFormInnerSum, m, A - m] // Together
```

(2) Out[14]=

```
0
```

Thus we get

(2) In[15]:=

```
innerSum == clFormInnerSum
```

(2) Out[15]=

$$\sum_{k=0}^{m}\cdot\left(\binom{m}{k}\cdot\right)^2 S[1,\,k]^2 == -\frac{(2\,m)!.\left(\sum_{i_1=1}^{m}\cdot\ \frac{1}{i_1^2}\right)}{4\,(m!.)^2}+\frac{9\,(2\,m)!.\left(\sum_{i_1=1}^{m}\cdot\ \frac{1}{i_1}\right)^2}{4\,(m!.)^2}-\frac{(2\,m)!.\left(\sum_{i_1=1}^{m}\cdot\ \frac{1}{(-1+2\,i_1)^2}\right)}{(m!.)^2}-$$

$$\frac{3\,(2\,m)!.\left(\sum_{i_1=1}^{m}\cdot\ \frac{1}{i_1}\right)\left(\sum_{i_1=1}^{m}\cdot\ \frac{1}{-1+2\,i_1}\right)}{(m!.)^2}+\frac{(2\,m)!.\left(\sum_{i_1=1}^{m}\cdot\ \frac{1}{-1+2\,i_1}\right)^2}{(m!.)^2}+\frac{3\,(2\,m)!.\left(\sum_{i_1=1}^{m}\cdot\ \frac{(i_1!.)^2}{i_1^2\,(2\,i_1)!.}\right)}{(m!.)^2}$$

for all m>=0 since both sides satisfy the same recurrence and agree with their first initial values.

Dealing with the outer sum

Finally, we focus on the outer sum where the summand equals

(2) In[16]:=

```
outerSummand = mySum[[1]] /. innerSum → clFormInnerSum
```

(2) Out[16]=

$$\frac{1}{(1+m)^2\binom{2m}{m}}\left(-\frac{(2m)!.\left(\sum_{i_1=1}^{m}\frac{1}{i_1^2}\right)}{4\,(m!.)^2}+\frac{9\,(2m)!.\left(\sum_{i_1=1}^{m}\frac{1}{i_1}\right)^2}{4\,(m!.)^2}-\frac{(2m)!.\left(\sum_{i_1=1}^{m}\frac{1}{(-1+2i_1)^2}\right)}{(m!.)^2}-\right.$$

$$\left.\frac{3\,(2m)!.\left(\sum_{i_1=1}^{m}\frac{1}{i_1}\right)\left(\sum_{i_1=1}^{m}\frac{1}{-1+2i_1}\right)}{(m!.)^2}+\frac{(2m)!.\left(\sum_{i_1=1}^{m}\frac{1}{-1+2i_1}\right)^2}{(m!.)^2}+\frac{3\,(2m)!.\left(\sum_{i_1=1}^{m}\frac{(i_1!.)^2}{i_1^2\,(2i_1)!.}\right)}{(m!.)^2}\right)$$

Note that here the central binomial coefficient cancels; this can be done automatically by the following call:

(2) In[17]:=

```
outerSummand = SigmaReduce[outerSummand, m]
```

(2) Out[17]=

$$\frac{\left(\sum_{i_1=1}^{m}\frac{1}{1-2i_1}\right)^2}{(1+m)^2}-\frac{\sum_{i_1=1}^{m}\frac{1}{i_1^2}}{4\,(1+m)^2}+\frac{3\left(\sum_{i_1=1}^{m}\frac{1}{1-2i_1}\right)\left(\sum_{i_1=1}^{m}\frac{1}{i_1}\right)}{(1+m)^2}+\frac{9\left(\sum_{i_1=1}^{m}\frac{1}{i_1}\right)^2}{4\,(1+m)^2}-\frac{\sum_{i_1=1}^{m}\frac{1}{(-1+2i_1)^2}}{(1+m)^2}+\frac{3\left(\sum_{i_1=1}^{m}\frac{(i_1!.)^2}{i_1^2\,(2i_1)!.}\right)}{(1+m)^2}$$

In other words, the truncated version of the double sum is

(2) In[18]:=

```
doubleSum = SigmaSum[outerSummand, {m, 0, A}]
```

(2) Out[18]=

$$\sum_{m=0}^{A}\left(\frac{\left(\sum_{i_1=1}^{m}\frac{1}{1-2i_1}\right)^2}{(1+m)^2}-\frac{\sum_{i_1=1}^{m}\frac{1}{i_1^2}}{4\,(1+m)^2}+\frac{3\left(\sum_{i_1=1}^{m}\frac{1}{1-2i_1}\right)\left(\sum_{i_1=1}^{m}\frac{1}{i_1}\right)}{(1+m)^2}+\frac{9\left(\sum_{i_1=1}^{m}\frac{1}{i_1}\right)^2}{4\,(1+m)^2}-\frac{\sum_{i_1=1}^{m}\frac{1}{(-1+2i_1)^2}}{(1+m)^2}+\frac{3\left(\sum_{i_1=1}^{m}\frac{(i_1!.)^2}{i_1^2\,(2i_1)!.}\right)}{(1+m)^2}\right)$$

which we manipulate as follows.

Telescoping

Using Sigma's telescoping solver, we obtain the simplification

(2) In[19]:=

```
resA = SigmaReduce[doubleSum];
resA = CollectProdSum[resA, 3]
```

(2) Out[20]=

$$16\left(\sum_{i_1=1}^{A}\cdot\frac{1}{1-2\,i_1}\right)-\frac{(1+2\,A)(3+2\,A)\left(\sum_{i_1=1}^{A}\cdot\frac{1}{1-2\,i_1}\right)^2}{(1+A)^2}+\frac{19}{8}\left(\sum_{i_1=1}^{A}\cdot\frac{1}{i_1^4}\right)+3\left(\sum_{i_1=1}^{A}\cdot\frac{1}{i_1^3}\right)+$$

$$\frac{(19+40\,A+20\,A^2)\left(\sum_{i_1=1}^{A}\cdot\frac{1}{i_1^2}\right)}{4\,(1+A)^2}-\frac{1}{8}\left(\sum_{i_1=1}^{A}\cdot\frac{1}{i_1^2}\right)^2+8\left(\sum_{i_1=1}^{A}\cdot\frac{1}{i_1}\right)-\frac{3\,(1+2\,A)(3+2\,A)\left(\sum_{i_1=1}^{A}\cdot\frac{1}{1-2\,i_1}\right)\left(\sum_{i_1=1}^{A}\cdot\frac{1}{i_1}\right)}{(1+A)^2}-$$

$$\frac{3\,(1+8\,A+4\,A^2)\left(\sum_{i_1=1}^{A}\cdot\frac{1}{i_1}\right)^2}{4\,(1+A)^2}+\frac{(1+2\,A)(3+2\,A)\left(\sum_{i_1=1}^{A}\cdot\frac{1}{(-1+2\,i_1)^2}\right)}{(1+A)^2}+\frac{3\left(\sum_{i_1=1}^{A}\cdot\frac{(i_1!.)^2}{i_1^2\,(2\,i_1)!.}\right)}{(1+A)^2}+3\left(\sum_{i_1=1}^{A}\cdot\frac{1}{i_1^2}\right)\left(\sum_{i_1=1}^{A}\cdot\frac{(i_1!.)^2}{i_1^2\,(2\,i_1)!.}\right)-$$

$$3\left(\sum_{i_1=1}^{A}\cdot\frac{\sum_{i_2=1}^{i_1}\cdot\frac{1}{1-2\,i_2}}{i_1^3}\right)-2\left(\sum_{i_1=1}^{A}\cdot\frac{\sum_{i_2=1}^{i_1}\cdot\frac{1}{1-2\,i_2}}{i_1^2}\right)+8\left(\sum_{i_1=1}^{A}\cdot\frac{\sum_{i_2=1}^{i_1}\cdot\frac{1}{1-2\,i_2}}{i_1}\right)+\sum_{i_1=1}^{A}\cdot\frac{\left(\sum_{i_2=1}^{i_1}\cdot\frac{1}{1-2\,i_2}\right)^2}{i_1^2}-3\left(\sum_{i_1=1}^{A}\cdot\frac{(i_1!.)^2\left(\sum_{i_2=1}^{i_1}\cdot\frac{1}{i_2^2}\right)}{i_1^2\,(2\,i_1)!.}\right)-$$

$$\frac{9}{2}\left(\sum_{i_1=1}^{A}\cdot\frac{\sum_{i_2=1}^{i_1}\cdot\frac{1}{i_2}}{i_1^3}\right)-3\left(\sum_{i_1=1}^{A}\cdot\frac{\sum_{i_2=1}^{i_1}\cdot\frac{1}{i_2}}{i_1^2}\right)+3\left(\sum_{i_1=1}^{A}\cdot\frac{\left(\sum_{i_2=1}^{i_1}\cdot\frac{1}{1-2\,i_2}\right)\left(\sum_{i_2=1}^{i_1}\cdot\frac{1}{i_2}\right)}{i_1^2}\right)+\frac{9}{4}\left(\sum_{i_1=1}^{A}\cdot\frac{\left(\sum_{i_2=1}^{i_1}\cdot\frac{1}{i_2}\right)^2}{i_1^2}\right)-\sum_{i_1=1}^{A}\cdot\frac{\sum_{i_2=1}^{i_1}\cdot\frac{1}{(-1+2\,i_2)^2}}{i_1^2}$$

The correctness can be checked immediately by the result. More precisely, let f[m] be the outerSummand and define g[m] as resA where A is replaces by m-1:

(2) In[21]:=

```
g = NShift[resA, A, -1] /. A → m // CollectProdSum
```

(2) Out[21]=

$$16\left(\sum_{i_1=1}^{m}\cdot\frac{1}{1-2\,i_1}\right)-4\left(\sum_{i_1=1}^{m}\cdot\frac{1}{1-2\,i_1}\right)^2+\frac{19}{8}\left(\sum_{i_1=1}^{m}\cdot\frac{1}{i_1^4}\right)+3\left(\sum_{i_1=1}^{m}\cdot\frac{1}{i_1^3}\right)-\frac{1}{8}\left(\sum_{i_1=1}^{m}\cdot\frac{1}{i_1^2}\right)^2+$$

$$\left(8-12\left(\sum_{i_1=1}^{m}\cdot\frac{1}{1-2\,i_1}\right)\right)\left(\sum_{i_1=1}^{m}\cdot\frac{1}{i_1}\right)-3\left(\sum_{i_1=1}^{m}\cdot\frac{1}{i_1}\right)^2+4\left(\sum_{i_1=1}^{m}\cdot\frac{1}{(-1+2\,i_1)^2}\right)+\left(\sum_{i_1=1}^{m}\cdot\frac{1}{i_1^2}\right)\left(5+3\left(\sum_{i_1=1}^{m}\cdot\frac{(i_1!.)^2}{i_1^2\,(2\,i_1)!.}\right)\right)-$$

$$3\left(\sum_{i_1=1}^{m}\cdot\frac{\sum_{i_2=1}^{i_1}\cdot\frac{1}{1-2\,i_2}}{i_1^3}\right)-2\left(\sum_{i_1=1}^{m}\cdot\frac{\sum_{i_2=1}^{i_1}\cdot\frac{1}{1-2\,i_2}}{i_1^2}\right)+8\left(\sum_{i_1=1}^{m}\cdot\frac{\sum_{i_2=1}^{i_1}\cdot\frac{1}{1-2\,i_2}}{i_1}\right)+\sum_{i_1=1}^{m}\cdot\frac{\left(\sum_{i_2=1}^{i_1}\cdot\frac{1}{1-2\,i_2}\right)^2}{i_1^2}-3\left(\sum_{i_1=1}^{m}\cdot\frac{(i_1!.)^2\left(\sum_{i_2=1}^{i_1}\cdot\frac{1}{i_2^2}\right)}{i_1^2\,(2\,i_1)!.}\right)-$$

$$\frac{9}{2}\left(\sum_{i_1=1}^{m}\cdot\frac{\sum_{i_2=1}^{i_1}\cdot\frac{1}{i_2}}{i_1^3}\right)-3\left(\sum_{i_1=1}^{m}\cdot\frac{\sum_{i_2=1}^{i_1}\cdot\frac{1}{i_2}}{i_1^2}\right)+3\left(\sum_{i_1=1}^{m}\cdot\frac{\left(\sum_{i_2=1}^{i_1}\cdot\frac{1}{1-2\,i_2}\right)\left(\sum_{i_2=1}^{i_1}\cdot\frac{1}{i_2}\right)}{i_1^2}\right)+\frac{9}{4}\left(\sum_{i_1=1}^{m}\cdot\frac{\left(\sum_{i_2=1}^{i_1}\cdot\frac{1}{i_2}\right)^2}{i_1^2}\right)-\sum_{i_1=1}^{m}\cdot\frac{\sum_{i_2=1}^{i_1}\cdot\frac{1}{(-1+2\,i_2)^2}}{i_1^2}$$

Then one can verify that the telescoping equation

g[m + 1] - g[m] == f[m] (**)

holds for all m>=0 by

(2) In[22]:=

```
NShift[g, m] - g - outerSummand // Together
```

(2) Out[22]=

```
0
```

Summing (**) over m from 0 to A yields resA which holds for all A>=0.

Performing the limit

To complete the calculation from above, we perform the limit A->Infinity as follows:

(2) In[23]:=

```
resLimit = SigmaLimit[resA, {}, A] /. LinInf → Infinity
```

(2) Out[23]=

$$20\,\text{li4half} - 16\,\ln 2 - 12\,\ln 2^2 + \frac{5\,\ln 2^4}{6} + \left(10 - 5\,\ln 2^2\right) z2 - \frac{38\,z2^2}{5} + \left(\frac{1}{2} + \frac{35\,\ln 2}{2}\right) z3 + 3\,z2 \left(\sum_{i_1=1}^{\infty}{}^{\cdot} \frac{(i_1!.)^2}{i_1^2\,(2\,i_1)!.}\right) - 3 \left(\sum_{i_1=1}^{\infty}{}^{\cdot} \frac{(i_1!.)^2 \left(\sum_{i_2=1}^{i_1}{}^{\cdot} \frac{1}{i_2^2}\right)}{i_1^2\,(2\,i_1)!.}\right)$$

which agrees with the output of the command EvaluateMultiSum:

(2) In[24]:=

```
resLimit - resInfiniteBinSum /. Sum[1/i2^2, {i2, 1, i1}] → S[2, i1] // Together
```

(2) Out[24]=

```
0
```

We give further details how the limit is calculated.

Details

First, we look at the sums with binomial contributions:

(2) In[25]:=

```
resAProd = Select[resA, ContainsProductExtension]
```

(2) Out[25]=

$$\frac{3 \left(\sum_{i_1=1}^{A}{}^{\cdot} \frac{(i_1!.)^2}{i_1^2\,(2\,i_1)!.}\right)}{(1 + A)^2} + 3 \left(\sum_{i_1=1}^{A}{}^{\cdot} \frac{1}{i_1^2}\right) \left(\sum_{i_1=1}^{A}{}^{\cdot} \frac{(i_1!.)^2}{i_1^2\,(2\,i_1)!.}\right) - 3 \left(\sum_{i_1=1}^{A}{}^{\cdot} \frac{(i_1!.)^2 \left(\sum_{i_2=1}^{i_1}{}^{\cdot} \frac{1}{i_2^2}\right)}{i_1^2\,(2\,i_1)!.}\right)$$

By the quotient rule one sees immediately that all finite binomials sums converge when sending A to Infinity. This leads to:

(2) In[26]:=

```
resProd = SigmaLimit[resAProd, {}, A] /. LinInf → Infinity
```

(2) Out[26]=

$$3\,z2\left(\sum_{i_1=1}^{\infty}\cdot\frac{(i_1!.)^2}{i_1^2\,(2\,i_1)!.}\right)-3\left(\sum_{i_1=1}^{\infty}\cdot\frac{(i_1!.)^2\left(\sum_{i_2=1}^{i_1}\cdot\frac{1}{i_2^2}\right)}{i_1^2\,(2\,i_1)!.}\right)$$

What remains open is the expression

(2) In[27]:=

```
resAProdFree = resA - resAProd;
```

which is built by sums being free of binomials:

(2) In[28]:=

```
var = Complement[Variables[resAProdFree], {A}]
```

(2) Out[28]=

$$\left\{\sum_{i_1=1}^{A}\cdot\frac{1}{1-2\,i_1},\ \sum_{i_1=1}^{A}\cdot\frac{1}{i_1^4},\ \sum_{i_1=1}^{A}\cdot\frac{1}{i_1^3},\ \sum_{i_1=1}^{A}\cdot\frac{1}{i_1^2},\ \sum_{i_1=1}^{A}\cdot\frac{1}{i_1},\ \sum_{i_1=1}^{A}\cdot\frac{1}{\left(-1+2\,i_1\right)^2},\ \sum_{i_1=1}^{A}\cdot\frac{\sum_{i_2=1}^{i_1}\cdot\frac{1}{1-2\,i_2}}{i_1^3},\ \sum_{i_1=1}^{A}\cdot\frac{\sum_{i_2=1}^{i_1}\cdot\frac{1}{1-2\,i_2}}{i_1^2},\ \sum_{i_1=1}^{A}\cdot\frac{\sum_{i_2=1}^{i_1}\cdot\frac{1}{1-2\,i_2}}{i_1},\right.$$

$$\left.\sum_{i_1=1}^{A}\cdot\frac{\left(\sum_{i_2=1}^{i_1}\cdot\frac{1}{1-2\,i_2}\right)^2}{i_1^2},\ \sum_{i_1=1}^{A}\cdot\frac{\sum_{i_2=1}^{i_1}\cdot\frac{1}{i_2}}{i_1^3},\ \sum_{i_1=1}^{A}\cdot\frac{\sum_{i_2=1}^{i_1}\cdot\frac{1}{i_2}}{i_1^2},\ \sum_{i_1=1}^{A}\cdot\frac{\left(\sum_{i_2=1}^{i_1}\cdot\frac{1}{1-2\,i_2}\right)\left(\sum_{i_2=1}^{i_1}\cdot\frac{1}{i_2}\right)}{i_1^2},\ \sum_{i_1=1}^{A}\cdot\frac{\left(\sum_{i_2=1}^{i_1}\cdot\frac{1}{i_2}\right)^2}{i_1^2},\ \sum_{i_1=1}^{A}\cdot\frac{\sum_{i_2=1}^{i_1}\cdot\frac{1}{\left(-1+2\,i_2\right)^2}}{i_1^2}\right\}$$

To handle the limit properly, we compute for each sum its asymptotic expansion using the HarmonicSums command SExpansion. In addition, we reduce the occurring constants with the HarmonicSums-command ReduceConstants. For details see

J. Ablinger, J. Blümlein and C. Schneider. Harmonic Sums and Polylogarithms Generated by Cyclotomic Polynomials. J. Math. Phys. 52 (2011) 102301. arXiv:1105.6063 [math-ph].

(2) In[29]:=

```
varExp = SExpansion[var, A, 2];
varExp = ReduceConstants[ varExp, ToKnownConstants → Automatic, Online → Automatic];
substExp = MapThread[Rule, {var, varExp}];
 substExp // MatrixForm[#, TableAlignments → Left] &
```

(2) Out[32]//MatrixForm=

$$\begin{pmatrix}
\sum_{i_1=1}^{A}{}' \frac{1}{1-2\,i_1} \to -\frac{1}{48\,A^2} - \ln 2 - \frac{LG[A]}{2} \\
\sum_{i_1=1}^{A}{}' \frac{1}{i_1^4} \to \frac{2\,z2^2}{5} \\
\sum_{i_1=1}^{A}{}' \frac{1}{i_1^3} \to -\frac{1}{2\,A^2} + z3 \\
\sum_{i_1=1}^{A}{}' \frac{1}{i_1^2} \to \frac{1}{2\,A^2} - \frac{1}{A} + z2 \\
\sum_{i_1=1}^{A}{}' \frac{1}{i_1} \to \frac{-1+6\,A}{12\,A^2} + LG[A] \\
\sum_{i_1=1}^{A}{}' \frac{1}{(-1+2\,i_1)^2} \to \frac{-1+3\,A\,z2}{4\,A} \\
\sum_{i_1=1}^{A}{}' \frac{\sum_{i_2=1}^{i_1}{}' \frac{1}{1-2\,i_2}}{i_1^3} \to -8\,li4half - \frac{\ln 2^4}{3} + \frac{1+4\,\ln 2}{8\,A^2} + 2\,\ln 2^2\,z2 + \frac{53\,z2^2}{20} - 7\,\ln 2\,z3 + \frac{LG[A]}{4\,A^2} \\
\sum_{i_1=1}^{A}{}' \frac{\sum_{i_2=1}^{i_1}{}' \frac{1}{1-2\,i_2}}{i_1^2} \to \frac{-2\,\ln 2+A\,(2+4\,\ln 2)-7\,A^2\,z3+(-1+2\,A)\,LG[A]}{4\,A^2} \\
\sum_{i_1=1}^{A}{}' \frac{\sum_{i_2=1}^{i_1}{}' \frac{1}{1-2\,i_2}}{i_1} \to \frac{-3+(8-48\,A)\,\ln 2-96\,A^2\,\ln 2^2+24\,A^2\,z2-4\,(-1+6\,A+24\,A^2\,\ln 2)\,LG[A]-24\,A^2\,LG[A]^2}{96\,A^2} \\
\sum_{i_1=1}^{A}{}' \frac{\left(\sum_{i_2=1}^{i_1}{}' \frac{1}{1-2\,i_2}\right)^2}{i_1^2} \to \frac{4\,\ln 2^2-4\,A\,(1+2\,\ln 2+2\,\ln 2^2)+9\,A^2\,z2^2-4\,(A-\ln 2+2\,A\,\ln 2)\,LG[A]+(1-2\,A)\,LG[A]^2}{8\,A^2} \\
\sum_{i_1=1}^{A}{}' \frac{\sum_{i_2=1}^{i_1}{}' \frac{1}{i_2}}{i_1^3} \to -\frac{1-2\,A^2\,z2^2+2\,LG[A]}{4\,A^2} \\
\sum_{i_1=1}^{A}{}' \frac{\sum_{i_2=1}^{i_1}{}' \frac{1}{i_2}}{i_1^2} \to \frac{-1-4\,A+8\,A^2\,z3+(2-4\,A)\,LG[A]}{4\,A^2} \\
\sum_{i_1=1}^{A}{}' \frac{\left(\sum_{i_2=1}^{i_1}{}' \frac{1}{1-2\,i_2}\right)\left(\sum_{i_2=1}^{i_1}{}' \frac{1}{i_2}\right)}{i_1^2} \to \frac{15+60\,\ln 2+240\,A\,(1+\ln 2)-8\,A^2\,(120\,li4half+5\,\ln 2^4-30\,\ln 2^2\,z2-6\,z2^2+105\,\ln 2\,z3)+30\,(1-4\,\ln 2+8\,A\,(1+\ln 2))\,LG[A]+}{240\,A^2} \\
\sum_{i_1=1}^{A}{}' \frac{\left(\sum_{i_2=1}^{i_1}{}' \frac{1}{i_2}\right)^2}{i_1^2} \to \frac{-5-40\,A+34\,A^2\,z2^2-10\,(1+4\,A)\,LG[A]+(10-20\,A)\,LG[A]^2}{20\,A^2} \\
\sum_{i_1=1}^{A}{}' \frac{\sum_{i_2=1}^{i_1}{}' \frac{1}{(-1+2\,i_2)^2}}{i_1^2} \to -8\,li4half - \frac{\ln 2^4}{3} - \frac{3\,z2}{4\,A} + 2\,\ln 2^2\,z2 + \frac{151\,z2^2}{40} + \frac{1+3\,z2}{8\,A^2} - 7\,\ln 2\,z3
\end{pmatrix}$$

Plugging the expansions into the expression yields:

(2) In[33]:=

```
resAProdFree = Collect[resAProdFree /. substExp /. LG[A] → Log[A] + EulerGamma, Log[A], Together]
```

(2) Out[33]=

$$\frac{1}{11\,520\,A^4\,(1+A)^2}\Big(-615+1520\,A+14\,260\,A^2-14\,400\,A^3-47\,520\,A^4-23\,040\,A^5-480\,A^2\,\text{EulerGamma}-43\,200\,A^4\,\text{EulerGamma}-23\,040\,A^5\,\text{EulerGamma}+5760\,A^2\,\text{EulerGamma}^2-5760\,A^4\,\text{EulerGamma}^2-11\,520\,A^5\,\text{EulerGamma}^2+230\,400\,A^4\,\text{li4half}+460\,800\,A^5\,\text{li4half}+230\,400\,A^6\,\text{li4half}+480\,A^2\,\text{ln2}-141\,120\,A^4\,\text{ln2}-345\,600\,A^5\,\text{ln2}-184\,320\,A^6\,\text{ln2}-11\,520\,A^2\,\text{EulerGamma}\,\text{ln2}+11\,520\,A^4\,\text{EulerGamma}\,\text{ln2}+23\,040\,A^5\,\text{EulerGamma}\,\text{ln2}+5760\,A^2\,\text{ln2}^2-144\,000\,A^4\,\text{ln2}^2-288\,000\,A^5\,\text{ln2}^2-138\,240\,A^6\,\text{ln2}^2+9600\,A^4\,\text{ln2}^4+19\,200\,A^5\,\text{ln2}^4+9600\,A^6\,\text{ln2}^4-5760\,A^2\,z2+120\,960\,A^4\,z2+241\,920\,A^5\,z2+115\,200\,A^6\,z2-57\,600\,A^4\,\text{ln2}^2\,z2-115\,200\,A^5\,\text{ln2}^2\,z2-57\,600\,A^6\,\text{ln2}^2\,z2-87\,552\,A^4\,z2^2-175\,104\,A^5\,z2^2-87\,552\,A^6\,z2^2+5760\,A^4\,z3+11\,520\,A^5\,z3+5760\,A^6\,z3+201\,600\,A^4\,\text{ln2}\,z3+403\,200\,A^5\,\text{ln2}\,z3+201\,600\,A^6\,\text{ln2}\,z3\Big)+\frac{1}{24\,A^2\,(1+A)^2}\Big(-1-90\,A^2-48\,A^3+24\,\text{EulerGamma}-24\,A^2\,\text{EulerGamma}-48\,A^3\,\text{EulerGamma}-24\,\text{ln2}+24\,A^2\,\text{ln2}+48\,A^3\,\text{ln2}\Big)\,\text{Log}[A]+\frac{\left(1-A^2-2\,A^3\right)\text{Log}[A]^2}{2\,A^2\,(1+A)^2}$$

The order of the expansions are sufficiently high so that the limit is now immediate. Using even the Mathematica Limit command one gets:

(2) In[34]:=

```
resProdFree = Limit[resAProdFree, A → Infinity]
```

(2) Out[34]=

$$20\,\text{li4half}+\frac{5\,\text{ln2}^4}{6}+10\,z2-\frac{38\,z2^2}{5}-\text{ln2}^2\left(12+5\,z2\right)+\frac{z3}{2}+\text{ln2}\left(-16+\frac{35\,z3}{2}\right)$$

We check that the derived result agrees with the above calculation:

(2) In[35]:=

```
resProd + resProdFree - resLimit // Together
```

(2) Out[35]=

```
0
```

Evaluating the infinite binomial sums

What is left is the valuation of the two infinite binomials sums

(2) In[36]:=

```
varL = Select[Variables[resInfiniteBinSum], ContainsProductExtension]
```

(2) Out[36]=

$$\left\{\sum_{i_1=1}^{\infty}{}^{\cdot}\,\frac{(i_1!.)^2}{i_1^2\,(2\,i_1)!.},\ \sum_{i_1=1}^{\infty}{}^{\cdot}\,\frac{(i_1!.)^2\,S[2,\,i_1]}{i_1^2\,(2\,i_1)!.}\right\}$$

using the tools of HarmonicSums . To assist the calculations efficiently, the table

(2) In[37]:=

```
hrel = << HLogCyclo3tow7.m;
```

has been calculated in advance that reduces cyclotomic harmonic polylogarithms such as

(2) In[38]:=

```
ToHarmonicSumsIntegrate[H[{0, 0}, {3, 1}, 1]]
```

(2) Out[38]=

$$\int_0^1 \int_0^{\tau_1} \frac{\tau_2}{\tau_1 \left(1 + \tau_2 + \tau_2^2\right)} \, d\tau_2 \, d\tau_1$$

up to weight 7 to basis sums induced by the shuffle product and other known relations; for details we refer to

J. Ablinger Proving Two Conjectural Series for $\zeta(7)$ and Discovering More Series for $\zeta(7)$. In: Mathematical Aspects of Computer and Information Science, D. Slamanig, E. Tsigaridas, Z. Zafeirakopoulos (ed.), pp. 42-47. 2020. Springer International Publishing, arXiv:1908.06631 [math.CO].

Among them, the basis sums (remaining sums) can be simplified to the following constants:

(2) In[39]:=

```
constrel = {H[{3, 0}, 1] → π/(3 √3), H[{3, 1}, 1] → 1/18 (-√3 π + 9 Log[3]), H[{0, 0}, {3, 1}, 1] →
    -1 + 7 π^2/54 - CS[{{3, 1, 2}}, ∞], H[{0, 0}, {0, 0}, {1, 0}, 1] → -z3, H[{0, 0}, {3, 1}, {3, 1}, 1] →
    1 + π/(2 √3) - 14 π^3/(243 √3) - 4 z3/9 + 2 π CS[{{3, 1, 2}}, ∞]/(3 √3) + 1/9 CS[{{1, 0, 2}, {3, 1, 1}}, ∞] - Log[3]/2,
  H[{0, 0}, {0, 0}, {0, 0}, {3, 1}, 1] → -1 + 79 π^4/7290 - CS[{{3, 1, 4}}, ∞],
  H[{0, 0}, {0, 0}, {3, 1}, {3, 1}, 1] → π/(6 √3) - 4 π^2/27 - 79 π^4/29160 - 26 π z3/(81 √3) +
    2 CS[{{3, 1, 2}}, ∞] - 1/6 π^2 CS[{{3, 1, 2}}, ∞] + CS[{{3, 1, 2}}, ∞]^2 + 3/2 (1 + CS[{{3, 1, 4}}, ∞]) +
    1/27 CS[{{1, 0, 3}, {3, 1, 1}}, ∞] + Log[3]/2, H[{0, 0}, {0, 0}, {0, 0}, {0, 0}, {1, 0}, 1] → -z5,
  H[{0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {0, 0}, {1, 0}, 1] → -z7};
```

(2) In[40]:=

To automatize the simplification we will use the following command:

(2) In[41]:=

```
ProduceConstants[sumL_List] :=
 Module[{step, varInd},
  Table[
   step = GetDefinition[sumL[[kk]]];
   varInd = step[[2, 1]];
   step = ComputeGeneratingFunction[(step[[1]]) // ToMathematica, x, {varInd, 1, ∞}];
   step = step /. x → 1 // SpecialGLToH // Expand // GLToH;
   step = step /. hrel /. constrel // Expand;
   step = ReduceConstants[step, ToKnownConstants → True] // Expand;
   step,
   {kk, 1, Length[sumL]}
  ]
 ]
```

Applying it to the two binomial sums we get the simplification

(2) In[42]:=

```
redL = ProduceConstants[varL]
```

(2) Out[42]=

$$\left\{\frac{z2}{3}, \frac{28\, z2^2}{135}\right\}$$

(2) In[43]:=

```
More precisely, we can carry out the substition
```

(2) In[43]:=

```
substBinSum = Table[varL[[k]] → redL[[k]], {k, Length[varL]}]
```

(2) Out[43]=

$$\left\{\sum_{i_1=1}^{\infty}{}_{\cdot} \frac{(i_1!_{\cdot})^2}{i_1^2\,(2\,i_1)!_{\cdot}} \to \frac{z2}{3}, \sum_{i_1=1}^{\infty}{}_{\cdot} \frac{(i_1!_{\cdot})^2\, S[2, i_1]}{i_1^2\,(2\,i_1)!_{\cdot}} \to \frac{28\, z2^2}{135}\right\}$$

and get

(2) In[44]:=

```
resFinal = resInfiniteBinSum /. substBinSum // Simplify
```

(2) Out[44]=

$$20\, li4half + \frac{5\, ln2^4}{6} + 10\, z2 - \frac{65\, z2^2}{9} - ln2^2\,(12 + 5\, z2) + \frac{z3}{2} + ln2\left(-16 + \frac{35\, z3}{2}\right)$$

Details for the second sum

We give details for the second (more involved) binomial sum:

(2) In[45]:=

```
sum2 = varL[[2]]
```

(2) Out[45]=

$$\sum_{\iota_1=1}^{\infty}. \frac{(\iota_1!.)^2\ S[2,\ \iota_1]}{\iota_1^2\ (2\ \iota_1)!.}$$

We compute its generating function (sneaking in the factor x^(ι_1)):

(2) In[46]:=

```
genFu = ComputeGeneratingFunction[sum2[[1]], x, sum2[[2]]]
```

(2) Out[46]=

$$x\left(\frac{9216 - 48\,192\,x + 12\,928\,x^2 + 636\,x^3 - 1056\,x^4 + 312\,x^5 - 48\,x^6 + 3\,x^7}{18\,432} + \frac{1}{768\,x}\right.$$
$$\frac{\sqrt{4-x}\ \left(768\ \sqrt{x} - 160\,x^{3/2} - 56\,x^{5/2} + 36\,x^{7/2} - 10\,x^{9/2} + x^{11/2}\right) \mathrm{GL}\left[\left\{\sqrt{4-\mathrm{VarGL}}\ \sqrt{\mathrm{VarGL}}\right\}, x\right] +}{128\,x}$$
$$\frac{\left(48 + 16\,x - 20\,x^2 + 8\,x^3 - x^4\right) \mathrm{GL}\left[\left\{\sqrt{4-\mathrm{VarGL}}\ \sqrt{\mathrm{VarGL}},\ \sqrt{4-\mathrm{VarGL}}\ \sqrt{\mathrm{VarGL}}\right\}, x\right]}{8\,x} + \frac{1}{32\,x}$$
$$\frac{\sqrt{4-x}\ \left(2\ \sqrt{x} - x^{3/2}\right) \mathrm{GL}\left[\left\{\sqrt{4-\mathrm{VarGL}}\ \sqrt{\mathrm{VarGL}},\ \sqrt{4-\mathrm{VarGL}}\ \sqrt{\mathrm{VarGL}},\ \sqrt{4-\mathrm{VarGL}}\ \sqrt{\mathrm{VarGL}}\right\}, x\right] +}{4\,x} + \frac{1}{16\,x}$$
$$\left.\mathrm{GL}\left[\left\{\sqrt{4-\mathrm{VarGL}}\ \sqrt{\mathrm{VarGL}},\ \sqrt{4-\mathrm{VarGL}}\ \sqrt{\mathrm{VarGL}},\ \sqrt{4-\mathrm{VarGL}}\ \sqrt{\mathrm{VarGL}},\ \sqrt{4-\mathrm{VarGL}}\ \sqrt{\mathrm{VarGL}}\right\}, x\right]\right)$$

The power series representation is given in terms of GL-functions, a typical example is:

(2) In[47]:=

```
ToHarmonicSumsIntegrate[
 GL[{Sqrt[4 - VarGL] Sqrt[VarGL], Sqrt[4 - VarGL] Sqrt[VarGL], Sqrt[4 - VarGL] Sqrt[VarGL], Sqrt[4 - VarGL] Sqrt[VarGL]}, x]]
```

(2) Out[47]=

$$\int_0^x \int_0^{\tau_1} \int_0^{\tau_2} \int_0^{\tau_3} \sqrt{4-\tau_1}\ \sqrt{\tau_1}\ \sqrt{4-\tau_2}\ \sqrt{\tau_2}\ \sqrt{4-\tau_3}\ \sqrt{\tau_3}\ \sqrt{4-\tau_4}\ \sqrt{\tau_4}\ d\tau_4\, d\tau_3\, d\tau_2\, d\tau_1$$

Now we go back to the original sum by replacing x by 1. In this simple form the integrals can be rewritten in terms of cyclotomic harmonic polylogarithms:

(2) In[48]:=

```
HRepresentation = genFu /. x → 1 // SpecialGLToH // Expand // GLToH
```

(2) Out[48]=

$$\frac{9}{32}\,\mathrm{H}[\{3,\,0\},\,1]^2-\frac{9}{16}\,\mathrm{H}[\{3,\,0\},\,\{3,\,0\},\,1]-12\,\mathrm{H}[\{1,\,0\},\,\{1,\,0\},\,1]\times\mathrm{H}[\{3,\,0\},\,\{3,\,0\},\,1]+$$
$$6\,\mathrm{H}[\{1,\,0\},\,\{3,\,0\},\,1]\times\mathrm{H}[\{3,\,0\},\,\{3,\,0\},\,1]+12\,\mathrm{H}[\{1,\,0\},\,\{3,\,1\},\,1]\times\mathrm{H}[\{3,\,0\},\,\{3,\,0\},\,1]+$$
$$6\,\mathrm{H}[\{3,\,0\},\,\{1,\,0\},\,1]\times\mathrm{H}[\{3,\,0\},\,\{3,\,0\},\,1]-12\,\mathrm{H}[\{3,\,0\},\,\{3,\,0\},\,1]^2-$$
$$6\,\mathrm{H}[\{3,\,0\},\,\{3,\,0\},\,1]\times\mathrm{H}[\{3,\,0\},\,\{3,\,1\},\,1]+12\,\mathrm{H}[\{3,\,0\},\,\{3,\,0\},\,1]\times\mathrm{H}[\{3,\,1\},\,\{1,\,0\},\,1]-$$
$$6\,\mathrm{H}[\{3,\,0\},\,\{3,\,0\},\,1]\times\mathrm{H}[\{3,\,1\},\,\{3,\,0\},\,1]-12\,\mathrm{H}[\{3,\,0\},\,\{3,\,0\},\,1]\times\mathrm{H}[\{3,\,1\},\,\{3,\,1\},\,1]+$$
$$12\,\mathrm{H}[\{3,\,0\},\,1]\times\mathrm{H}[\{1,\,0\},\,\{1,\,0\},\,\{3,\,0\},\,1]-6\,\mathrm{H}[\{3,\,0\},\,1]\times\mathrm{H}[\{1,\,0\},\,\{3,\,0\},\,\{3,\,0\},\,1]-$$
$$12\,\mathrm{H}[\{3,\,0\},\,1]\times\mathrm{H}[\{1,\,0\},\,\{3,\,1\},\,\{3,\,0\},\,1]-6\,\mathrm{H}[\{3,\,0\},\,1]\times\mathrm{H}[\{3,\,0\},\,\{1,\,0\},\,\{3,\,0\},\,1]+$$
$$12\,\mathrm{H}[\{1,\,0\},\,1]\times\mathrm{H}[\{3,\,0\},\,\{3,\,0\},\,\{1,\,0\},\,1]-6\,\mathrm{H}[\{3,\,0\},\,1]\times\mathrm{H}[\{3,\,0\},\,\{3,\,0\},\,\{1,\,0\},\,1]-$$
$$12\,\mathrm{H}[\{3,\,1\},\,1]\times\mathrm{H}[\{3,\,0\},\,\{3,\,0\},\,\{1,\,0\},\,1]-6\,\mathrm{H}[\{1,\,0\},\,1]\times\mathrm{H}[\{3,\,0\},\,\{3,\,0\},\,\{3,\,0\},\,1]+$$
$$24\,\mathrm{H}[\{3,\,0\},\,1]\times\mathrm{H}[\{3,\,0\},\,\{3,\,0\},\,\{3,\,0\},\,1]+6\,\mathrm{H}[\{3,\,1\},\,1]\times\mathrm{H}[\{3,\,0\},\,\{3,\,0\},\,\{3,\,0\},\,1]-$$
$$12\,\mathrm{H}[\{1,\,0\},\,1]\times\mathrm{H}[\{3,\,0\},\,\{3,\,0\},\,\{3,\,1\},\,1]+6\,\mathrm{H}[\{3,\,0\},\,1]\times\mathrm{H}[\{3,\,0\},\,\{3,\,0\},\,\{3,\,1\},\,1]+$$
$$12\,\mathrm{H}[\{3,\,1\},\,1]\times\mathrm{H}[\{3,\,0\},\,\{3,\,0\},\,\{3,\,1\},\,1]+6\,\mathrm{H}[\{3,\,0\},\,1]\times\mathrm{H}[\{3,\,0\},\,\{3,\,1\},\,\{3,\,0\},\,1]-$$
$$12\,\mathrm{H}[\{3,\,0\},\,1]\times\mathrm{H}[\{3,\,1\},\,\{1,\,0\},\,\{3,\,0\},\,1]+6\,\mathrm{H}[\{3,\,0\},\,1]\times\mathrm{H}[\{3,\,1\},\,\{3,\,0\},\,\{3,\,0\},\,1]+$$
$$12\,\mathrm{H}[\{3,\,0\},\,1]\times\mathrm{H}[\{3,\,1\},\,\{3,\,1\},\,\{3,\,0\},\,1]-12\,\mathrm{H}[\{1,\,0\},\,\{1,\,0\},\,\{3,\,0\},\,\{3,\,0\},\,1]+$$
$$6\,\mathrm{H}[\{1,\,0\},\,\{3,\,0\},\,\{3,\,0\},\,\{3,\,0\},\,1]+12\,\mathrm{H}[\{1,\,0\},\,\{3,\,1\},\,\{3,\,0\},\,\{3,\,0\},\,1]+$$
$$6\,\mathrm{H}[\{3,\,0\},\,\{1,\,0\},\,\{3,\,0\},\,\{3,\,0\},\,1]-12\,\mathrm{H}[\{3,\,0\},\,\{3,\,0\},\,\{3,\,0\},\,\{3,\,0\},\,1]-$$
$$6\,\mathrm{H}[\{3,\,0\},\,\{3,\,1\},\,\{3,\,0\},\,\{3,\,0\},\,1]+12\,\mathrm{H}[\{3,\,1\},\,\{1,\,0\},\,\{3,\,0\},\,\{3,\,0\},\,1]-$$
$$6\,\mathrm{H}[\{3,\,1\},\,\{3,\,0\},\,\{3,\,0\},\,\{3,\,0\},\,1]-12\,\mathrm{H}[\{3,\,1\},\,\{3,\,1\},\,\{3,\,0\},\,\{3,\,0\},\,1]$$

A typical example is:

(2) In[49]:=

```
ToHarmonicSumsIntegrate[H[{3, 1}, {3, 1}, {3, 0}, {3, 0}, 1]]
```

(2) Out[49]=

$$\int_0^1\int_0^{\tau_1}\int_0^{\tau_2}\int_0^{\tau_3}\frac{\tau_1\,\tau_2}{(1+\tau_1+\tau_1^2)\,(1+\tau_2+\tau_2^2)\,(1+\tau_3+\tau_3^2)\,(1+\tau_4+\tau_4^2)}\,d\tau_4\,d\tau_3\,d\tau_2\,d\tau_1$$

Finally, we use the pre-calculated table of relations of cyclotomic polylogarithms and all the arising integrals collapse to the form

(2) In[50]:=

```
HRepresentation = HRepresentation /. hrel // Expand
```

(2) Out[50]=

$$\frac{21}{5}\,\mathrm{H}[\{3,\,0\},\,1]^4$$

Finally, we use the fact that the basis sums can be simplified further to

(2) In[51]:=

```
HRepresentation = HRepresentation /. constrel
```

(2) Out[51]=

$$\frac{7\,\pi^4}{1215}$$

which can be rewritten in terms of Zeta-values:

(2) In[52]:=

```
HRepresentation = ReduceConstants[HRepresentation, ToKnownConstants → True] // Expand
```

(2) Out[52]=

$$\frac{28\ z2^2}{135}$$